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Large-scale quantum chemistry applications with sample-based quantum diagonalization

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Two recent papers extending hybrid quantum algorithms to the biomolecular scale (Cleveland Clinic + MSU + IBM + RIKEN)

Molecular Quantum Computations on a Protein

Akhil Shajan, Danil Kaliakin, Fangchun Liang, Thaddeus Pellegrini, Hakan Doga, Subhamoy Bhowmik, Susanta Das, Antonio Mezzacapo, Mario Motta, and Kenneth M. Merz, Jr.*

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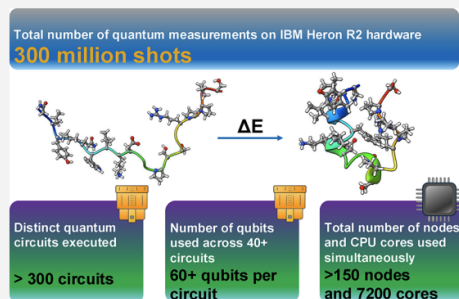
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ABSTRACT: We present the implementation of a fragment-based, quantum-centric supercomputing workflow for computing molecular electronic structure using quantum hardware. The workflow is applied to predict the relative energies of two conformers of the 303-atom Trp-cage miniprotein. The methodology employs wave function–based embedding (EWF) as the underlying fragmentation framework, in which all atoms in the system are explicitly included in the configuration interaction (CI) simulations. We employ sample-based quantum diagonalization (SQD) solver for challenging fragments and full configuration interaction (FCI) solver for trivial fragments. The EWF-(FCI,SQD) results are compared against EWF-MP2 and EWF-CCSD benchmarks. The impact of fragmentation on the predicted relative energies of the Trp-cage conformers is further evaluated by comparison with unfragmented RI-MP2 and DLPNO-CCSD calculations. The results demonstrate that large-scale electronic configuration interaction (CI) simulations of protein systems containing hundreds or even thousands of atoms can be realized through the combined use of quantum and classical computing resources.



Quantum Physics

[Submitted on 1 May 2026]

Crossing the 12,000-atom barrier with heterogeneous quantum-classical supercomputing: quantum chemistry of protein-ligand complexes

Kenneth M. Merz Jr., Akhil Shajan, Danil Kaliakin, Fangchun Liang, Yuichi Otsuka, Tomonori Shirakawa, Lukas Broers, Han Xu, Miwako Tsuji, Mitsuhsa Sato, Seiji Yunoki, Ryo Wakizaka, Yukio Kawashima, Jun Doi, Toshinari Itoko, Hiroshi Horii, Thaddeus Pellegrini, Javier Robledo Moreno, Kevin J. Sung, Ella Fejer, Robert Walkup, Seetharami Seelam, Mario Motta

Ab initio wavefunction methods provide accurate molecular simulations but their computational scaling restricts applications to small systems. We develop a workflow combining quantum embedding to decompose a molecule into fragments with a heterogeneous quantum-classical (HQC) method to simulate fragments. We sample fragment electronic configurations on two 156-qubit quantum processors (ibm_cleveland, ibm_kobe), using up to 94 qubits, running 9,200 circuits for over 100 hours, collecting $1.3 \cdot 10^9$ measurement outcomes – the most resource-intensive HQC computation for quantum chemistry to date. We compute fragment wavefunctions via optimized subspace diagonalization on two supercomputers (Fugaku, Miyabi-G), achieving 72.5% parallel efficiency with scalable distributed linear algebra kernels. We simulate two protein-ligand complexes spanning dispersion- and electrostatics-dominated regimes (11,608 and 12,635 atoms), demonstrate > 40x increase in system size and up to 210x improvement in accuracy over the previous state-of-the-art, with HQC matching coupled-cluster (CCSD) accuracy in fragment energies, and establish a scalable pathway for systematically improvable biomolecular simulations.

Comments: 12 pages, 6 figures, 9 tables

Subjects: **Quantum Physics (quant-ph)**; Chemical Physics (physics.chem-ph); Computational Physics (physics.comp-ph)

Cite as: [arXiv:2605.01138 \[quant-ph\]](#)

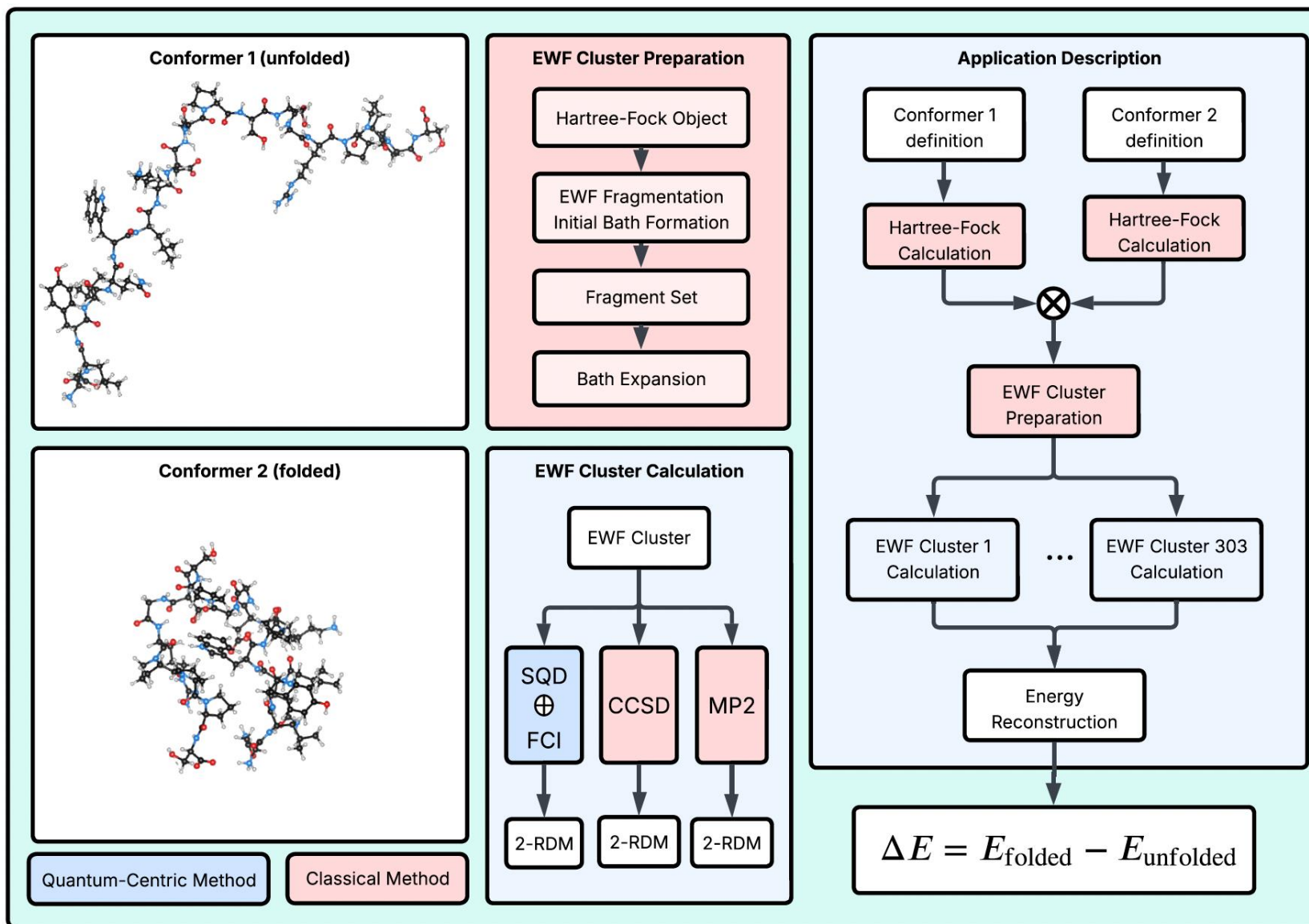
(or [arXiv:2605.01138v1 \[quant-ph\]](#) for this version)

<https://doi.org/10.48550/arXiv.2605.01138>

Introduction

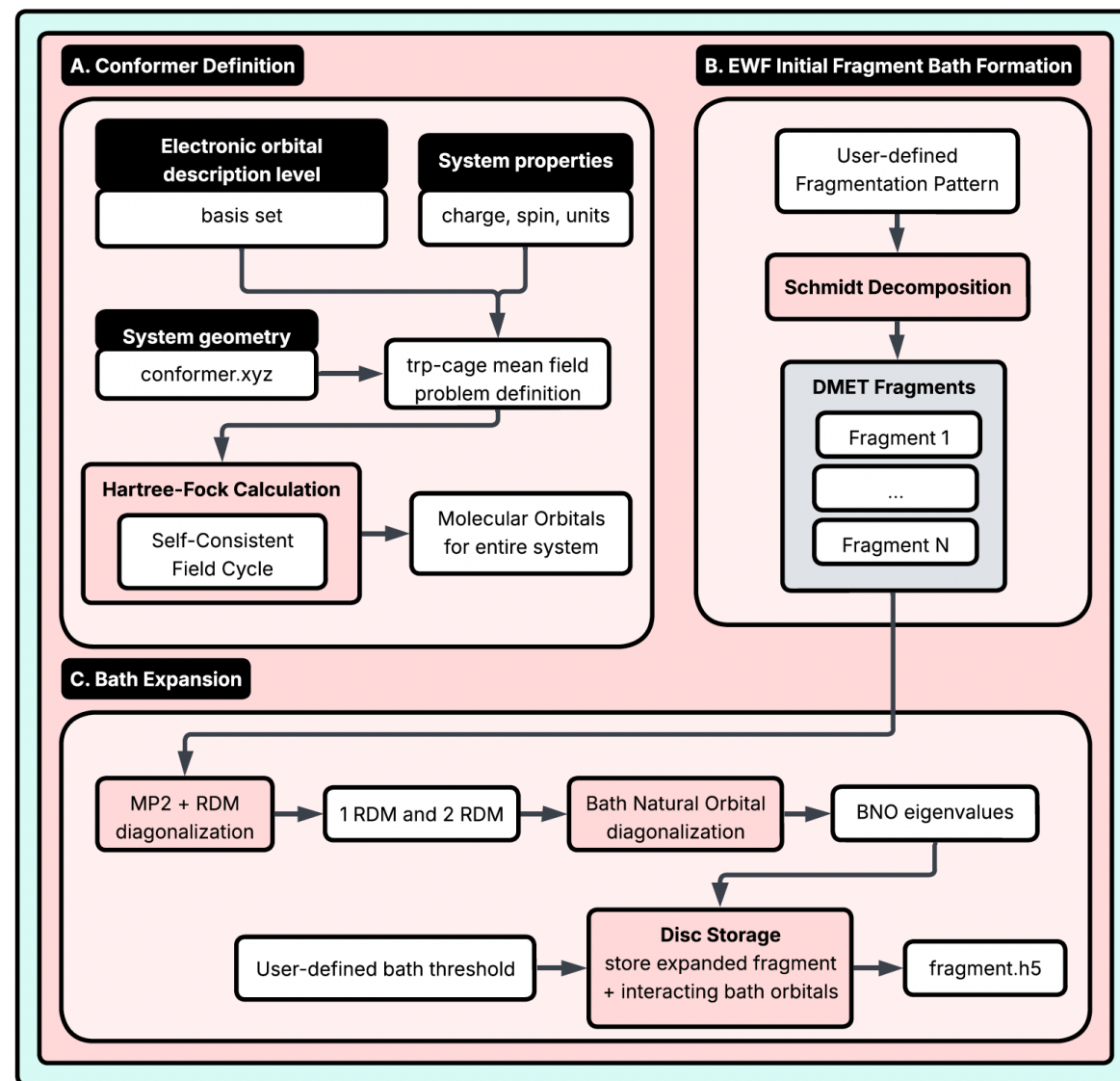
- Simulating the electronic structure of molecules and materials one of the most promising applications of quantum computing
- Quantum selected configuration interaction (QSCI) and sample-based quantum diagonalization have scaled up to 36 molecular orbital (MO) examples
- Scaling of current techniques to biomolecular scale requires embedding/fragment-based techniques
- Wavefunction-based embedding method (EWF) presents a non-iterative embedding scheme with atomic fragments
- Application to two conformers of a Trp-cage mini-protein (300 atoms)
- Fragments spanning 6-33 MOs
- Generate fragment clusters with EWF (classical), generate fragment properties with SQD (quantum+classical), and reconstruct total energy from fragments (classical)

Workflow Overview



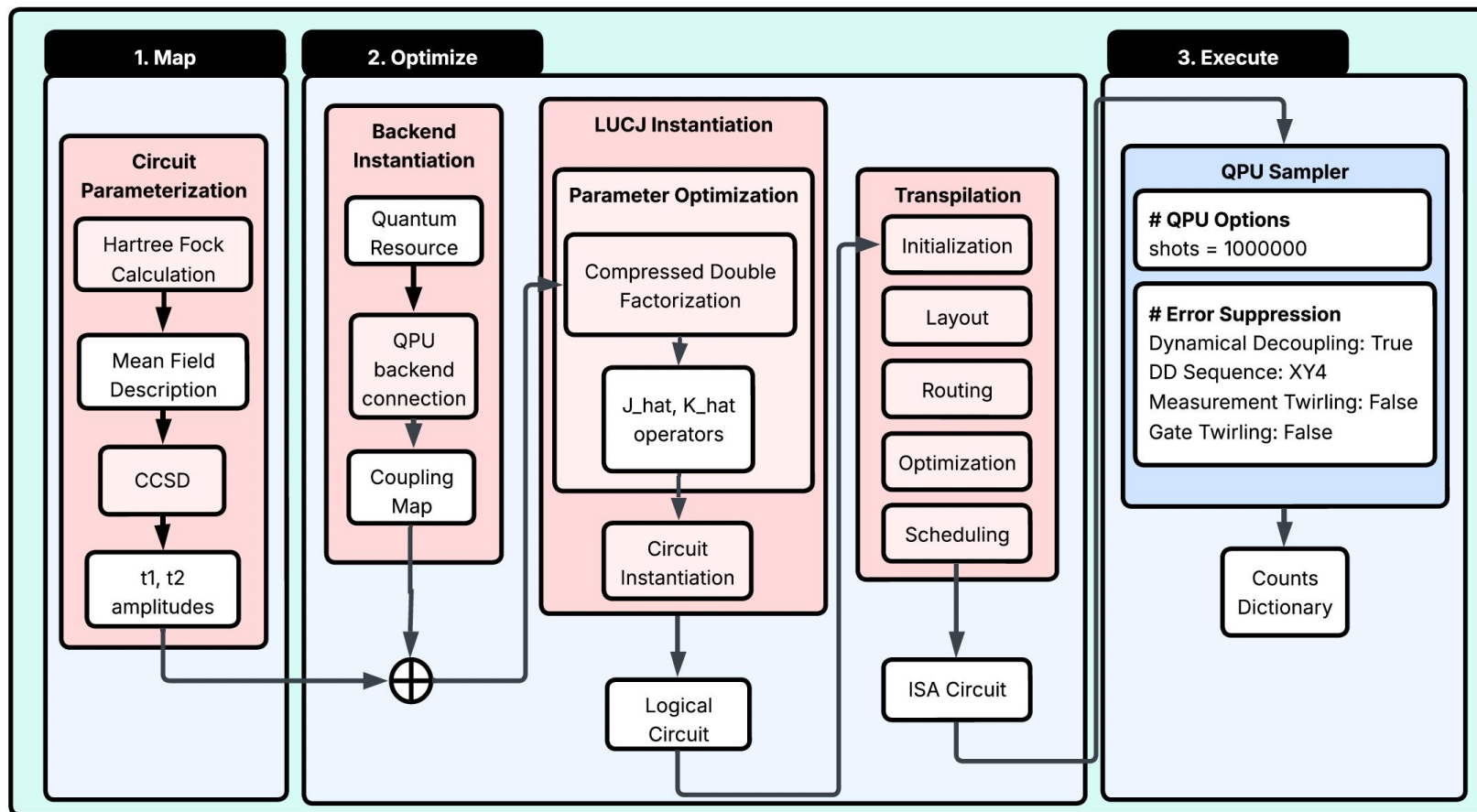
Methods: Embedded Wave Function (EWF)

- Inspired by density matrix embedding theory (DMET), fragment + bath approach
- Define fragments from full-system mean-field calculation and Schmidt decomposition (intrinsic atomic orbitals)
- Expand DMET bath using MP2 to define interacting bath orbitals
- Threshold η determines inclusion of bath orbitals
- Combined fragment and bath define EWF cluster



Methods: Sample-based quantum diagonalization (SQD) and extension (ext-SQD)

- Hybrid quantum-classical fragment solver
- Quantum hardware used to prepare and sample candidate electronic configurations
- Compact local unitary cluster Jastrow ansatz seeded with classical parameters
- Classical HPC used to perform error mitigation and subspace diagonalization
- Expansion of subspace basis by considering single excitations around dominant configurations



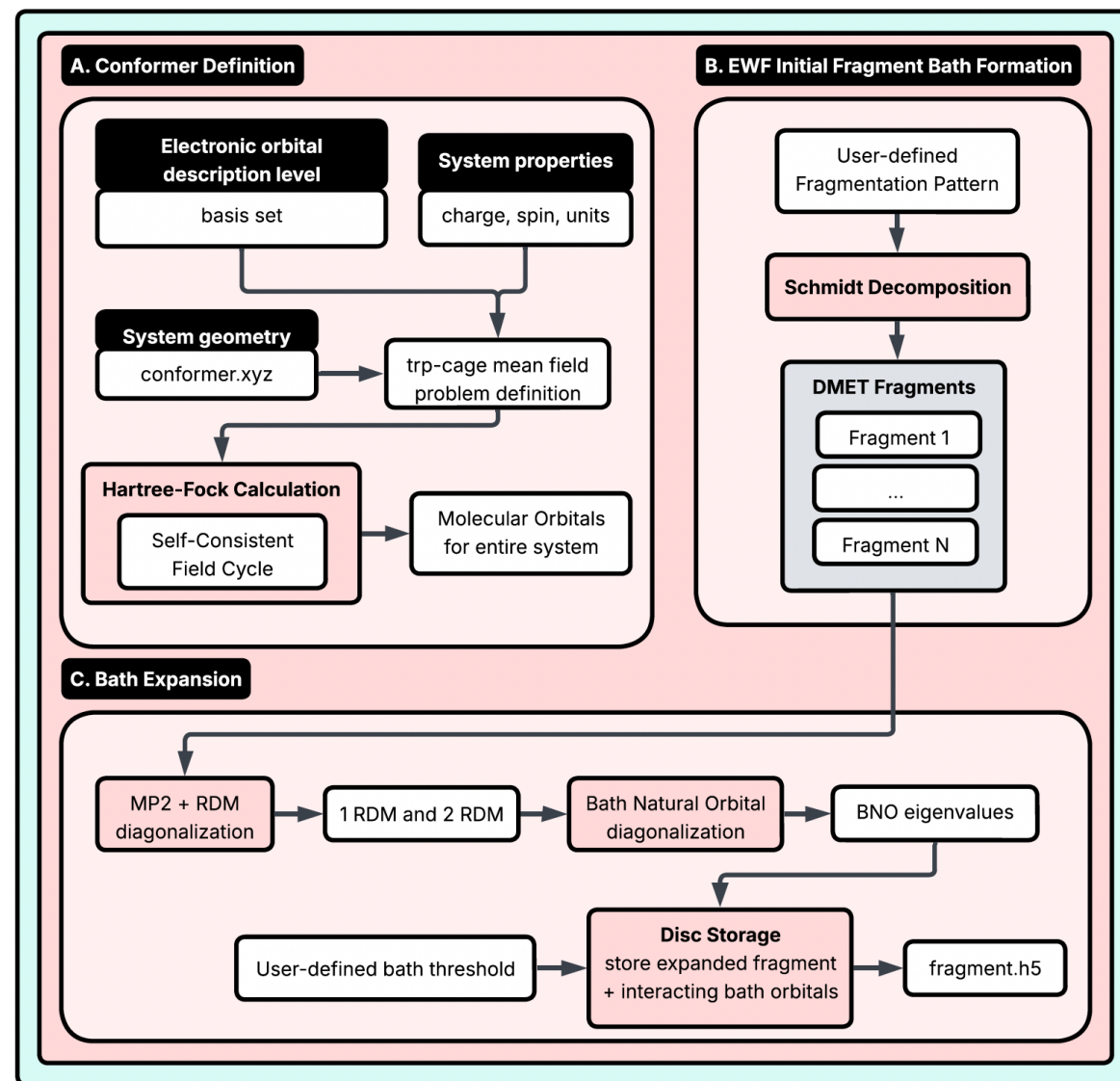
Computational Details: Classical preprocessing

Full-system HF calculations:

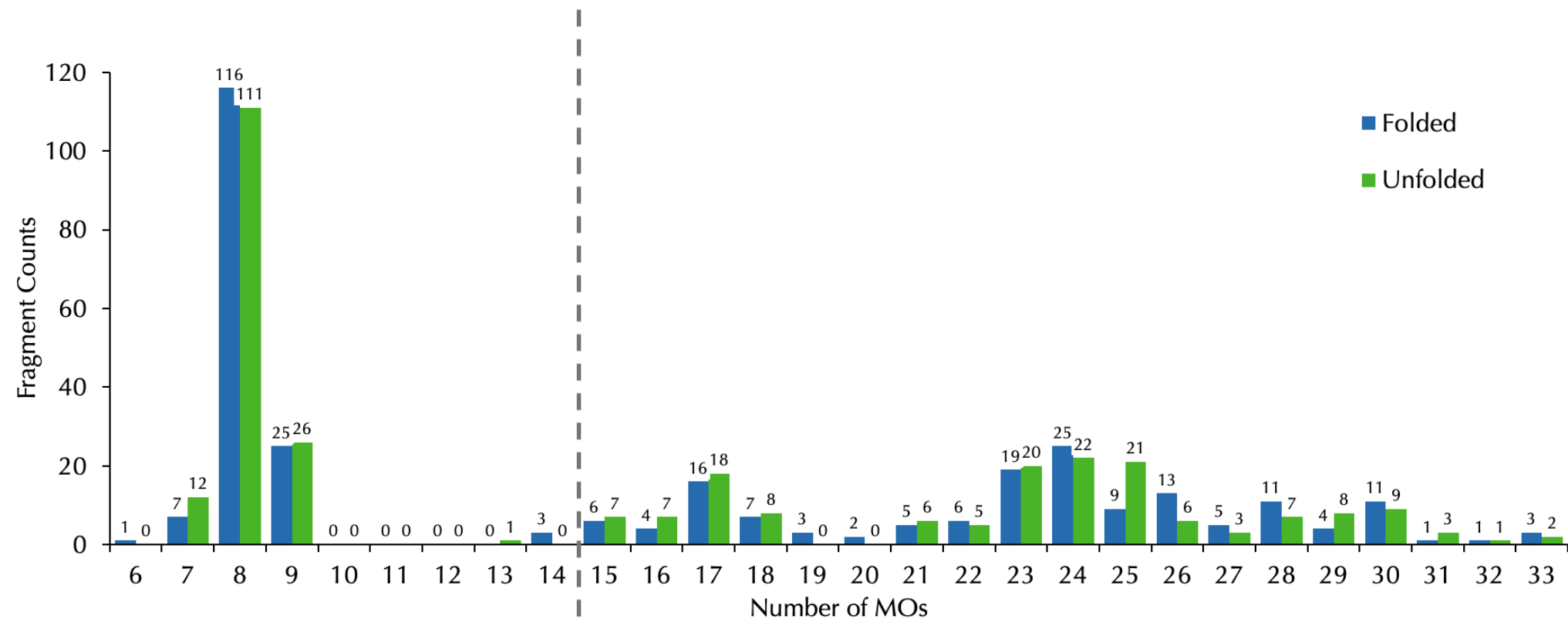
- HF calculations performed using GPU4PySCF package
- STO-3G basis + density fitting
- Mean field calculations performed on a NVIDIA V100S-PCIe 32GB accelerator paired with a 2.4 GHz Intel Xeon Platinum 8260 CPU
- ~4 min per conformer

EWF cluster formation:

- Interacting bath expansion using MP2 natural orbitals
- 197 GB RAM and 4 CPU cores, total of 487 GB of disk storage/conformer
- ~30 min wall time
- Write individual fragment data to .h5 files



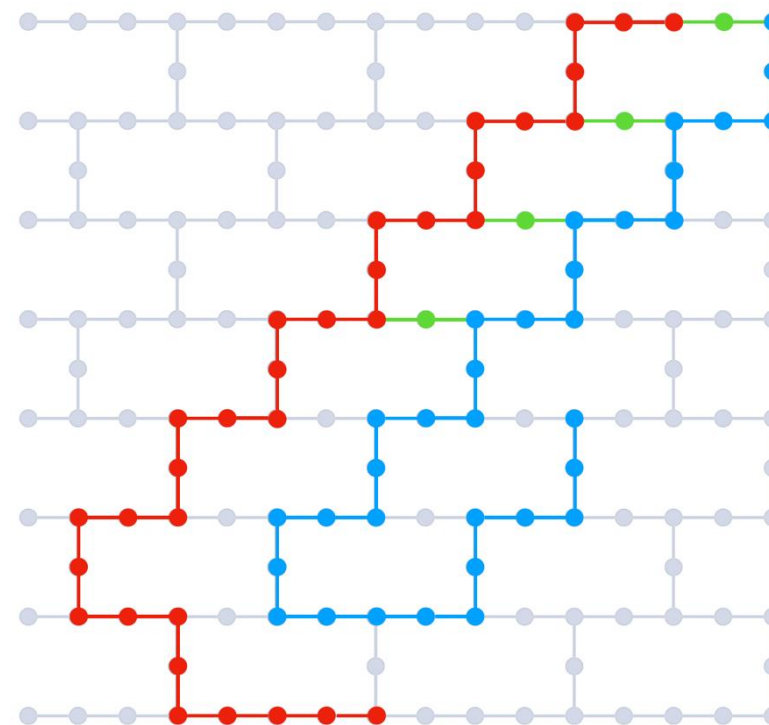
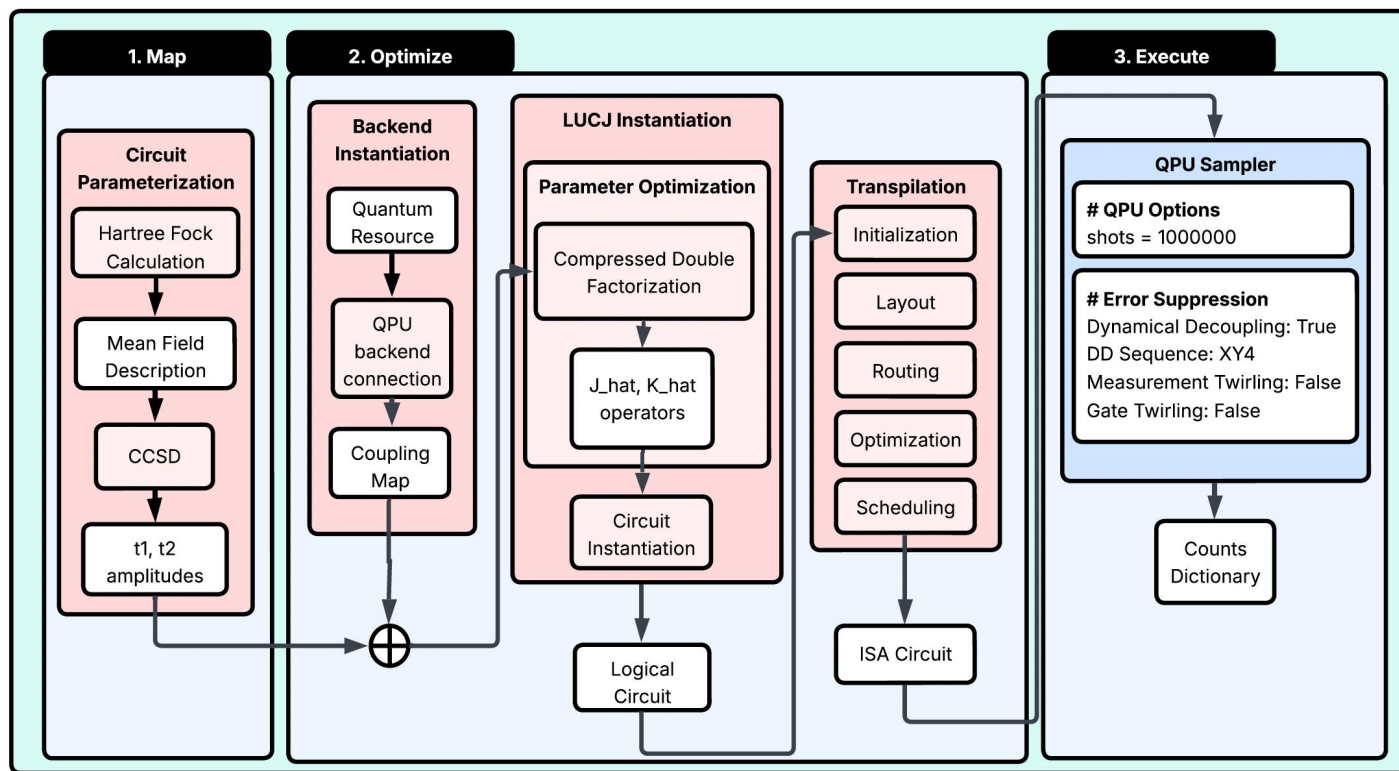
Fragmentation of Trp-conformers



Conformer	6-14 MO Fragments	15-33 MO Fragments	Total Fragments
Folded	152	151	303
Unfolded	150	153	303

Computational Details: Quantum Sampling

- Fragments >15 orbitals solved with classical exact diagonalization
- 15-33 orbital fragments solved using ext-SQD and classical reference (MP2 and CCSD)
- SQD sampling calculations performed using LUCJ ansatz with single layer
- 1M shots/fragment
- Executed on IBM Heron-R2 QPU (Heavy hex)



Computational Details: Configuration recovery and subspace diagonalization

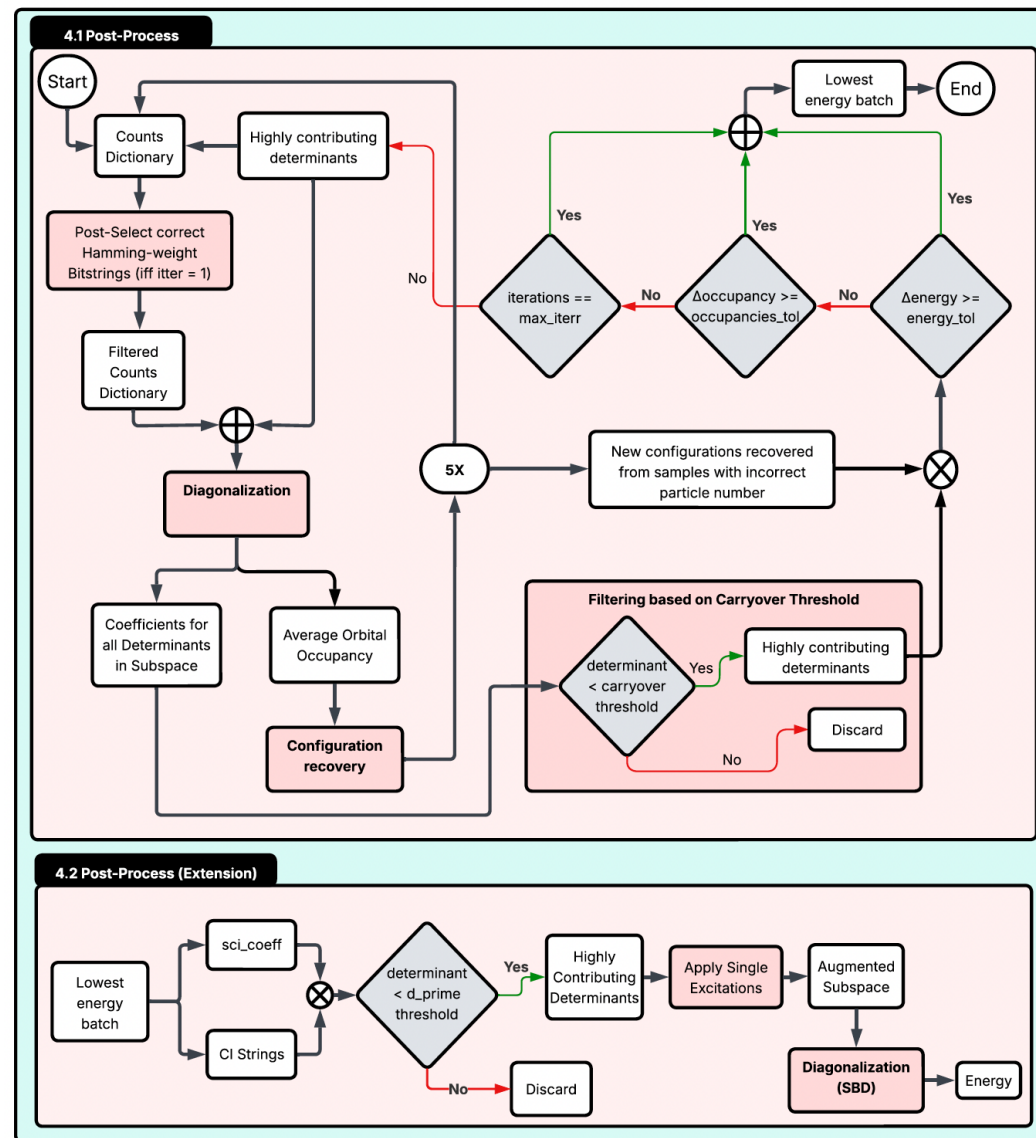
- Self-consistent configuration recovery error mitigation to restore proper particle number
- Most computationally intensive aspect of algorithm
- Asynchronous, parallelizable

15-24 MO fragments:

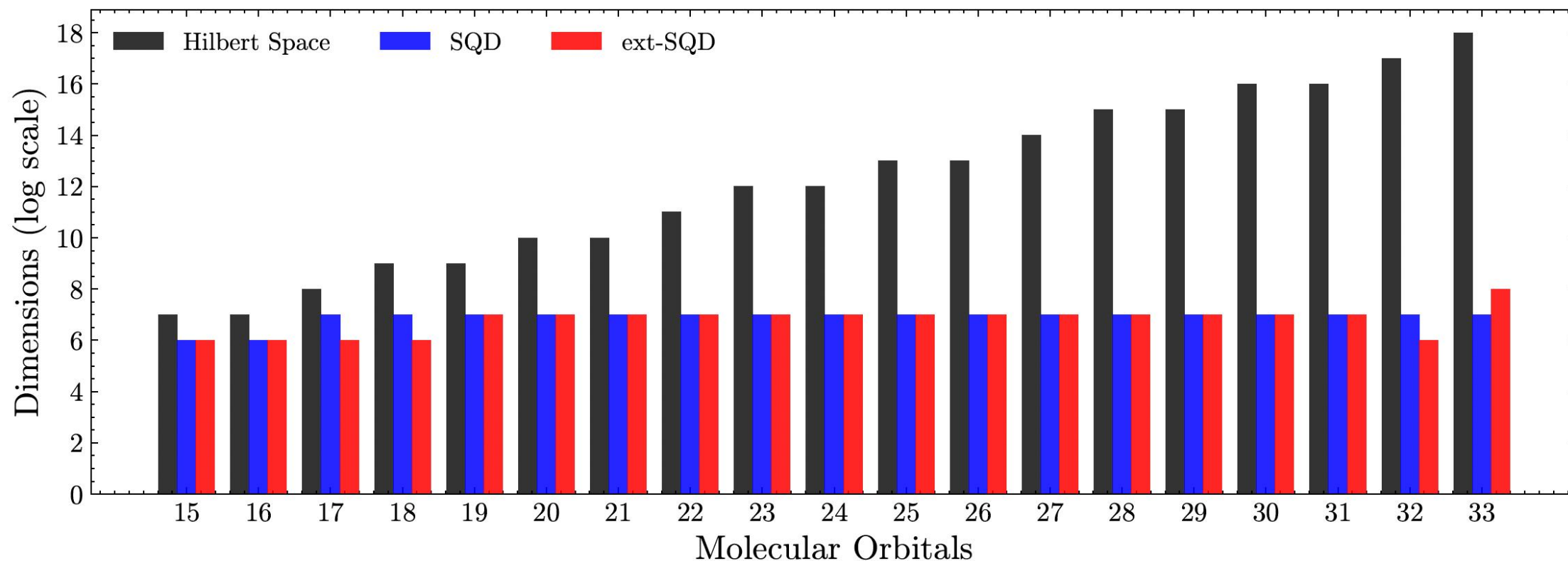
- Michigan State University HPC cluster,
- 80-100 nodes
- 493 GB RAM, 128 CPU

25-33 MO fragments:

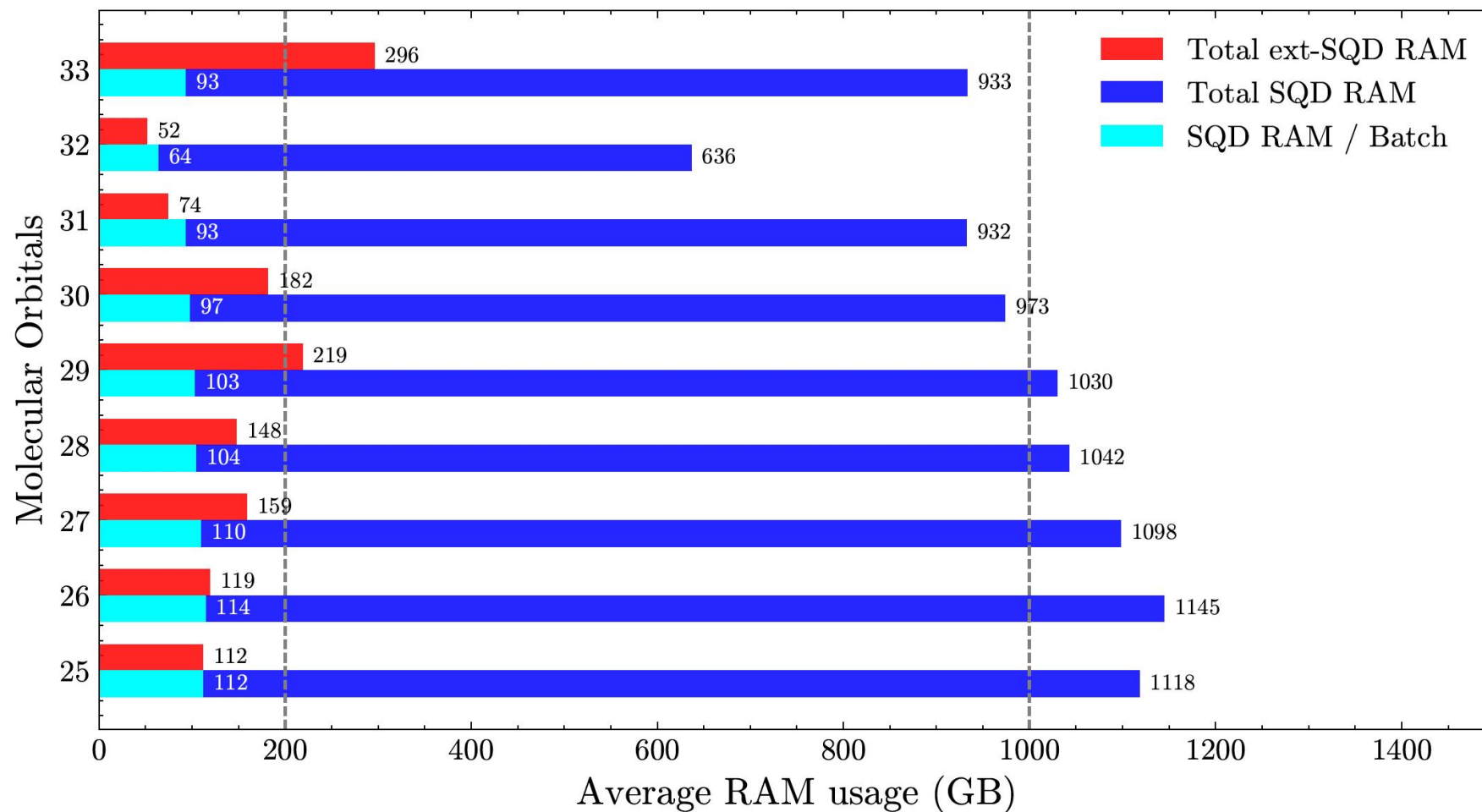
- Cleveland Clinic Foundation HPC
- 3 TB RAM, 96 CPU/6 TB RAM, 112 CPU
- All calcs utilized 48 CPU/node



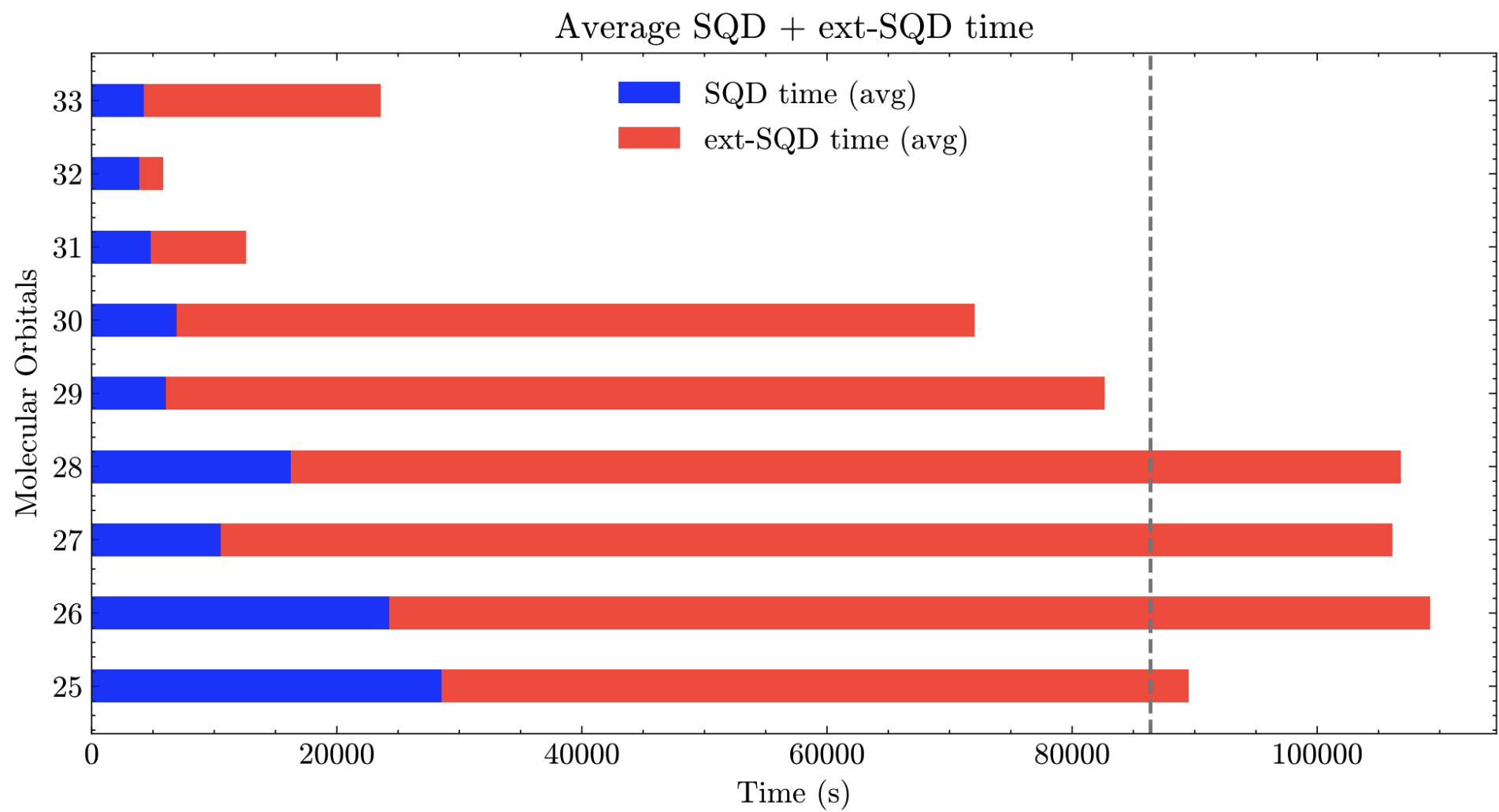
Hilbert Space Reduction with SQD and Ext-SQD



Memory requirements for SQD and Ext-SQD



Classical timings for SQD and Ext-SQD



Results and Conclusions

- Total energy reconstructed from 1- and 2-RDMs from each fragment projected and symmetrized in the full-system basis
- N-representable global expectation values
- Largest embedded CI calculations to date
- First example of biological-scale problem treated with quantum sampling

Method	$E_{\text{unfolded}}[\text{Ha}]$	$E_{\text{folded}}[\text{Ha}]$	$\Delta E [\text{kcal/mol}]$
EWF-MP2	-7352.7377	-7352.8327	59.61
EWF-CCSD	-7354.0653	-7354.1413	47.70
EWF-(FCI,SQD)	-7354.1372	-7354.2256	55.43

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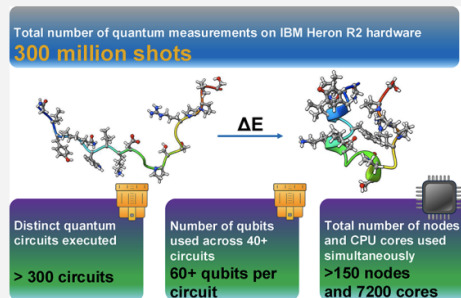
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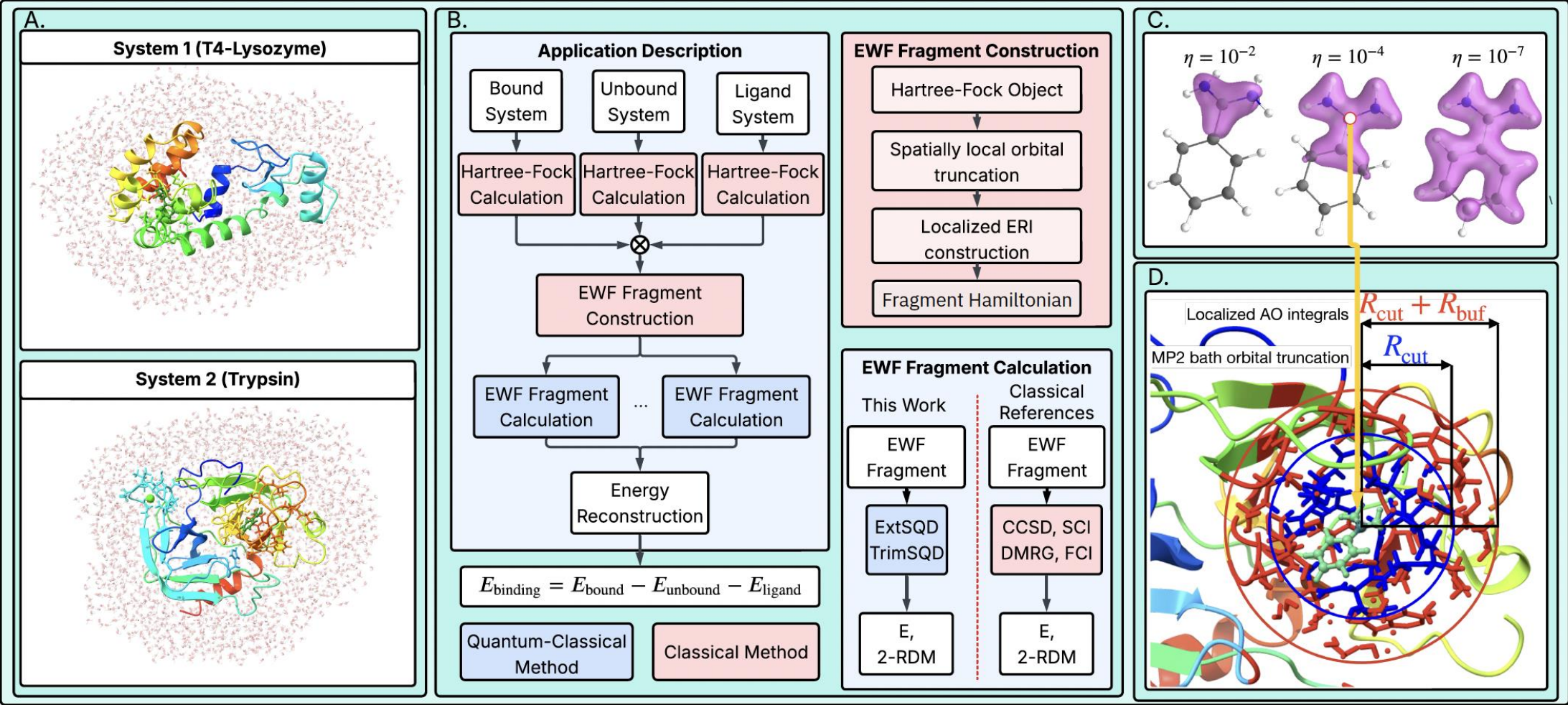
Introduction

- Determination of protein-ligand binding energies, important properties for drug discovery applications
- Two protein-ligand complexes (11,608 and 12,635 atoms)
- Two 156-qubit quantum (ibm_cleveland and ibm_kobe)
- Two leadership-class supercomputers Fugaku and Miyabi-G

Improvements over previous work:

- Quantum embedding method with $\mathcal{O}(1)$ scaling
- TrimSQD: an improved hybrid quantum classical fragment solver to deliver 210X improvement in accuracy
- SBD-G: a GPU accelerated selected-basis diagonalization subroutine
- Problem size 40X larger than previous state of the art (303 atoms)

Workflow Overview



Methods: Limitations of EWF and extensions

Traditional EWF Method

- Full system HF calculation and orbital localization at $O(N_{orb}^3)$
- Evaluation of electron repulsion integrals (ERI) for the full system at $O(N_{orb}^4)$ memory cost
- Full system MP2 calculation to generate correlated bath embedding at $O(N_{orb}^5)$

This Work

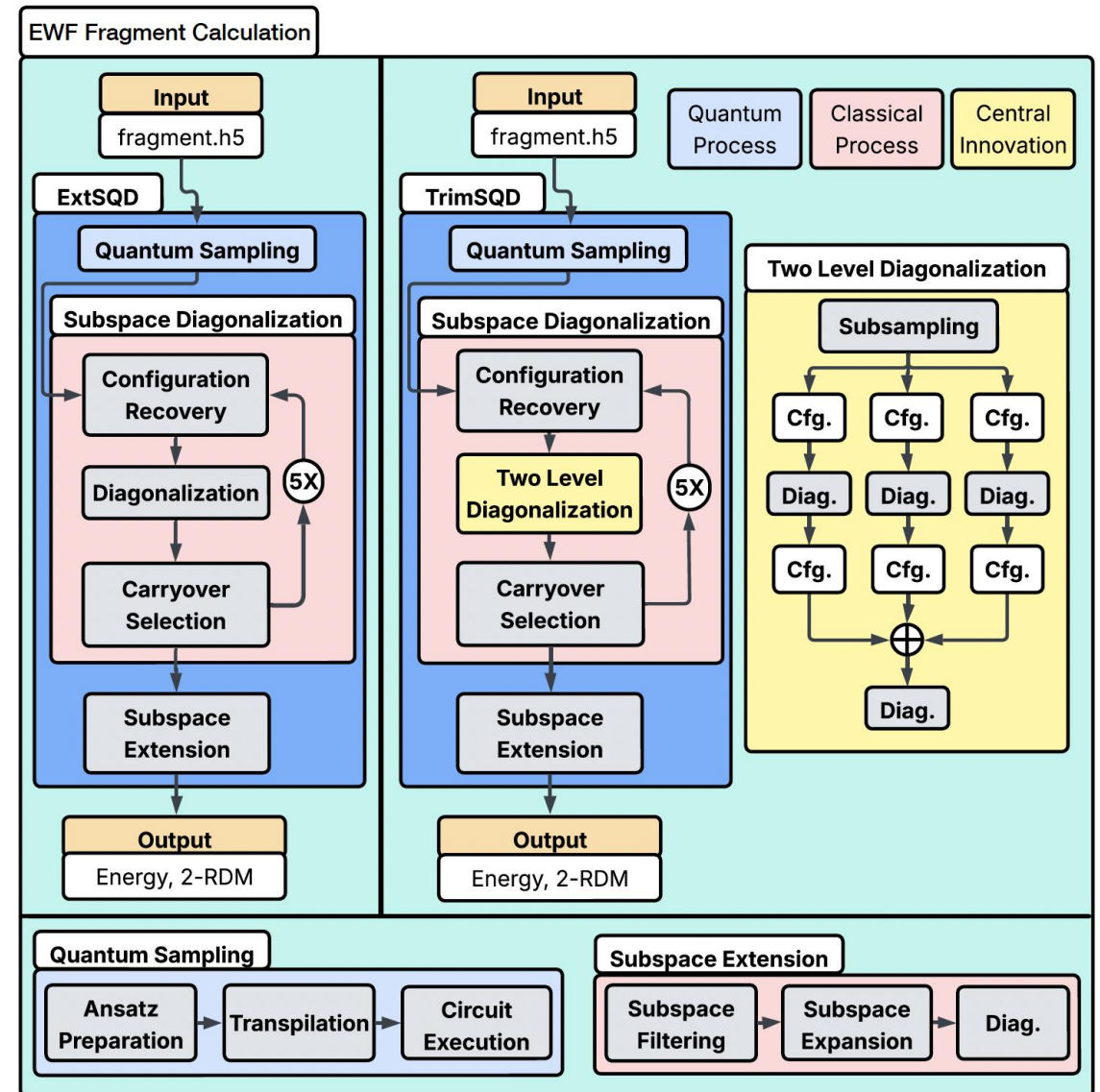
- Full system HF calculation and orbital localization at $O(N_{orb}^3)$
- **Spatially localized orbital truncation:** MP2 calculation to compute bath interactions for atomic fragment only include those spatially localized within a given cutoff radius ($R_{cut} = 7\text{\AA}$)
- **Localized electron repulsion integrals:** restrict construction of ERIs to only those within an additional buffer of the cutoff ($R_{buf} = 3\text{\AA}$)

Methods: TrimSQR fragment solver and SBD algorithm

- Introduces trimming of determinants to previous ExtSQR method to achieve higher accuracies
- Within the configuration recovery loop, batches of sampled configurations are used for subspace diagonalization and configurations with weight above a threshold k_1 are retained
- In TrimSQR, a second level of filtering of configuration is carried out with configuration weights above k_2 retained
- Configuration set is extended with all configurations one or two-particle excitations from the original set
- Final diagonalization over the extended subspace

Selected-Basis Diagonalization (SBD)

- In-house diagonalization code to accelerate parallel execution
- Distributed memory design
- GPU acceleration



Test systems

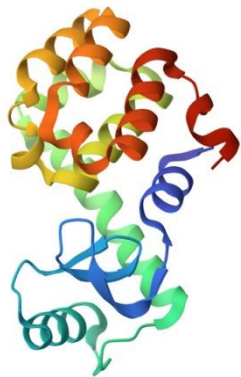
I) Trypsin-Benzamidine

- Serine protease enzyme and inhibitor
- 12,635 atoms; 31,795 MOs
- PDB ID: 3PTB



II) T4-Lysozyme—n-butyl-benzene

- Engineered protein and ligand with known high affinity
- 11,608 atoms; 28,844 MOs
- PDB ID: 4W57



System	MOs	Fragments	Sample size	QPU	QPU time per fragment	Total QPU time
I	13-26	4,296	100K	ibm_c	≈29 s	56 h
	27-45	549	500K	ibm_c	≈142 s	
II	13-18	751	100K	ibm_k	≈29 s	50 h
	19-26	3,228	100K	ibm_c	≈29 s	
	27-35	467	500K	ibm_c	≈142 s	

Quantum and HPC platforms

Quantum Resources

- Two 156-qubit Heron r2 QPUs
- ibm_cleveland (ibm_c) and ibm_kobe (ibm_k)
- 106 h total runtime
- 9,200 total quantum circuits (1,016 between 58-94 qubits)
- Largest circuits: 94 qubits, 246 2-qubit gate depth, 17 μ s execution time
- $1.3 \cdot 10^9$ shots total

Classical HPC Resources

Fugaku

- 158,976 nodes
- Fujitsu A64FX processor
- 48 compute cores, 32 GB high-bandwidth memory

Miyabi-G

- GPU-accelerated 1,120 NVIDIA GH200 Grace-Hopper Superchip
- 72-core NVIDIA Grace CPU/NVIDIA Hopper H100 GPU

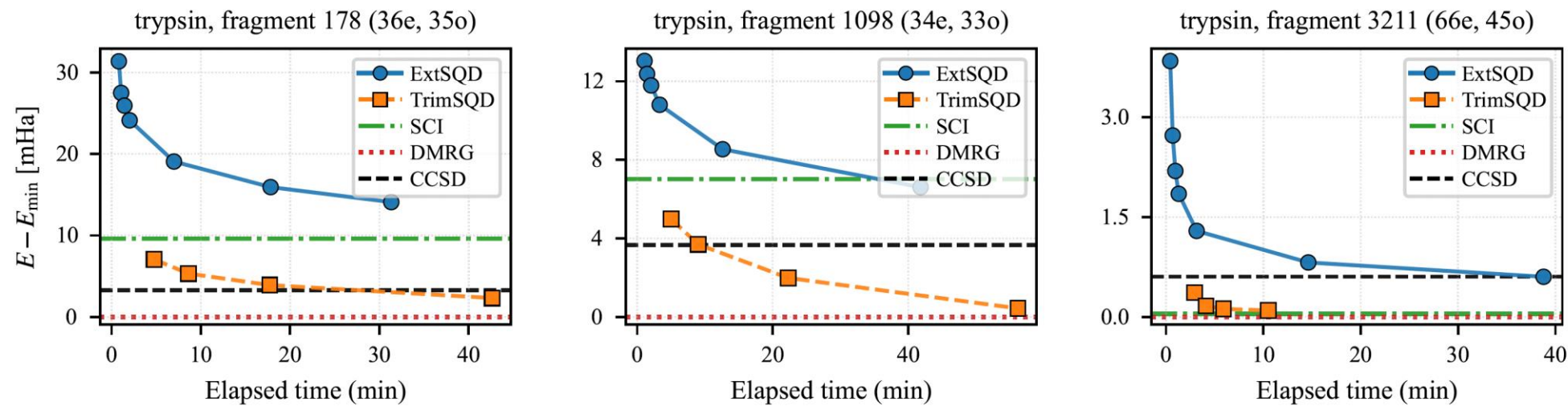
Results: Computational Timings

Step (Platform)	Time (node-hours)		Peak Nodes (% usage)
	Trypsin	T4-lys	
Quantum Sampling (ibm_c)	≈56	≈44	1/94 (100%/60%)
Quantum Sampling (ibm_k)	--	≈6	1/40(100%/25%)
Diagonalization (Fugaku)	369,315	308,528	152,064(95.7%)
Diagonalization (Miyabi-G)	4,281	3,455	1,104(98.6%)

- T4-lysozyme: 4.58 h on Fugaku (2.32 h work-bound runtime), 9.43 h on Miyabi-G (3.82 h work-bound runtime)
- Trypsin: 78.36 h (including substantial queueing time) on Fugaku (1.94 h work-bound runtime), 9.43 on Miyabi-G (3.08 h work-bound runtime)
- At $N_{node}=64$, improved SBD-G algorithm achieves parallel efficiency of 72.5%

Results: Binding Energies

System	Method	E_{bound} (Ha)	E_{unbound} (Ha)	E_{ligand} (Ha)	E_{binding} (kcal/mol)
Trypsin, benzamidine	EWF-CCSD	-319,413.0227	-319,038.0806	-374.9875	28.46
	EWF-TrimSQD	-319,415.8966	-319,040.9552	-374.9986	35.89
T4-Lysozyme, <i>n</i> -butyl-benzene	EWF-CCSD	-289,139.2838	-288,756.3865	-382.9139	10.45
	EWF-TrimSQD	-289,141.6090	-288,758.7087	-382.9152	9.30



Conclusions

- Exploiting localization allows scaling of EWF to over 12,000 atoms, exceeding previous studies by 40x
- TrimSQD utilizing a two-level filtering of configurations increases the accuracy over ExtSQD
- With 9,200 circuits and 1.3×10^9 measurements, most quantum resource-extensive quantum chemistry study to date
- Limitations on basis set and EWF do not predict binding, but good agreement with classical benchmarks
- GPU-enabled selected basis diagonalization routine enables massive parallelization on leadership computing
- In both studies, quantum and classical resources are asynchronous